

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1506

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I Ismail M (192524128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ismail M

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Ans:

A company experiences unpredictable traffic spikes during seasonal sales, such as Black Friday or Diwali. Cloud **auto-scaling** automatically increases or decreases the number of servers based on CPU usage, memory utilization, or the number of incoming requests. During peak traffic, additional virtual machines or containers are launched automatically. Once traffic reduces, unnecessary resources are removed, helping the company save costs.

Load balancing distributes incoming user requests evenly across multiple servers. This prevents any single server from becoming overloaded and ensures that all servers share the workload efficiently. If one server fails, the load balancer redirects traffic to healthy servers, ensuring continuous service availability.

This approach significantly **reduces downtime** because applications continue running even during sudden traffic spikes. Customers experience faster page loading, uninterrupted transactions, and a smoother shopping experience, leading to higher customer satisfaction and increased sales.

Monitoring tools such as AWS CloudWatch, Azure Monitor, or Google Cloud Monitoring continuously track system metrics like CPU utilization, memory usage, response time, and network traffic. These tools automatically trigger scaling actions whenever predefined thresholds are exceeded and also generate alerts if any failures occur.

Real-world example: During the Amazon Prime Day sale, millions of users access the website simultaneously. Amazon uses cloud auto-scaling and load balancing to dynamically allocate additional computing resources, ensuring that customers can browse products, add items to carts, and complete purchases without delays or service interruptions.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.

How can encryption and access control protect sensitive backup data?

Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Ans:

Organizations store large amounts of backup data using cloud storage services because they provide high durability and reliability. Cloud providers store multiple copies of data across different physical servers and geographical locations. This **data redundancy** ensures that even if one storage device or data center fails, the backup remains available from another location.

Cloud storage provides **high durability**, often reaching 99.999999999% (11 nines), by continuously replicating data and performing automatic integrity checks.

Sensitive backup data is protected using **encryption**. Data is encrypted before transmission (using SSL/TLS) and while stored in the cloud (AES-256 encryption). Even if unauthorized users gain access to storage devices, encrypted data cannot be read without the correct encryption keys.

Access control is implemented using Identity and Access Management (IAM), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC). Only authorized users can access or modify backup files.

Compared to traditional storage systems, cloud storage offers significant **cost advantages**:

- No need to purchase expensive storage hardware.
- Reduced maintenance and electricity costs.
- Pay only for the storage actually used.
- Automatic scaling eliminates overprovisioning.

Disaster Recovery Example: A company's primary data center is destroyed due to flooding. Since daily backups are stored in the cloud, the organization restores all business data to a new server within hours, allowing business operations to continue with minimal data loss.

3. **A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.**

Ans:

A mobile application deployed globally serves users from many countries. A **Content Delivery Network (CDN)** improves performance by storing copies of frequently accessed content at multiple edge locations around the world. Instead of downloading data from the original server, users receive content from the nearest CDN server.

This greatly **reduces latency**, resulting in faster loading times, smoother application performance, and reduced network congestion.

Edge servers are strategically located close to end users. These servers cache images, videos, application files, and web pages. When a user accesses the application, the nearest edge server responds immediately instead of contacting the distant origin server.

Cloud providers such as AWS CloudFront, Azure CDN, and Google Cloud CDN support **global availability** through worldwide networks of edge servers, automatic failover mechanisms, load balancing, and multiple regional data centers. Even if one region experiences problems, traffic is redirected to another healthy region.

Real-world example: Netflix uses CDNs to stream movies and TV shows. When a user in India watches a movie, the video is delivered from the nearest CDN edge server rather than directly from servers located in another country. This provides smooth streaming with minimal buffering.

4. **A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.**

Ans:

Financial institutions handle highly confidential customer information and must comply with regulations such as GDPR, PCI DSS, ISO 27001, and local banking regulations. Cloud providers support these requirements by offering certified infrastructure, compliance reports, and regular security audits.

Logging records every user activity, login attempt, file access, and administrative action. Audit logs help organizations investigate security incidents and demonstrate regulatory compliance.

Monitoring continuously observes cloud resources, network activity, and security events. Suspicious activities generate real-time alerts, allowing administrators to respond quickly.

Encryption protects sensitive financial information both while it is transmitted across networks and while stored in cloud databases. Encryption ensures that intercepted data remains unreadable without the correct decryption keys.

To satisfy **data sovereignty** requirements, organizations can choose cloud regions within their own country so customer data never leaves national borders. Cloud providers also allow customers to define data residency policies.

Example: A bank processes online fund transfers. Customer account information, transaction details, and payment records are encrypted before storage. Every transaction is logged for auditing purposes, and monitoring systems detect suspicious login attempts or fraudulent transactions. All customer data remains stored within the country's approved cloud region to satisfy financial regulations.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Ans:

Containers package an application along with its required libraries and dependencies. Unlike **Virtual Machines (VMs)**, containers do not include a complete guest operating system. Instead, they share the host operating system kernel.

As a result:

- Containers start within seconds.
- They consume much less memory and storage.
- Multiple containers can run on the same server efficiently.
- Performance is faster because there is less overhead.

Virtual Machines require an entire operating system for each instance, making them heavier and slower to start.

Kubernetes simplifies deployment by automatically managing containers. It handles:

- Automatic deployment of applications.
- Scaling containers based on workload.
- Load balancing between containers.
- Automatic restart of failed containers.
- Rolling updates with minimal downtime.
- Service discovery and resource management.

Containers also provide excellent **portability**. A Docker container developed on a developer's laptop runs the same way on AWS, Microsoft Azure, Google Cloud Platform, or an on-premises server without modification.

Example: An online shopping application built using microservices contains separate containers for:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Order Management Service
- Notification Service

Kubernetes manages each microservice independently, automatically scaling the payment service during heavy shopping periods while keeping other services unchanged. This improves performance, reliability, and resource utilization.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight

Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity